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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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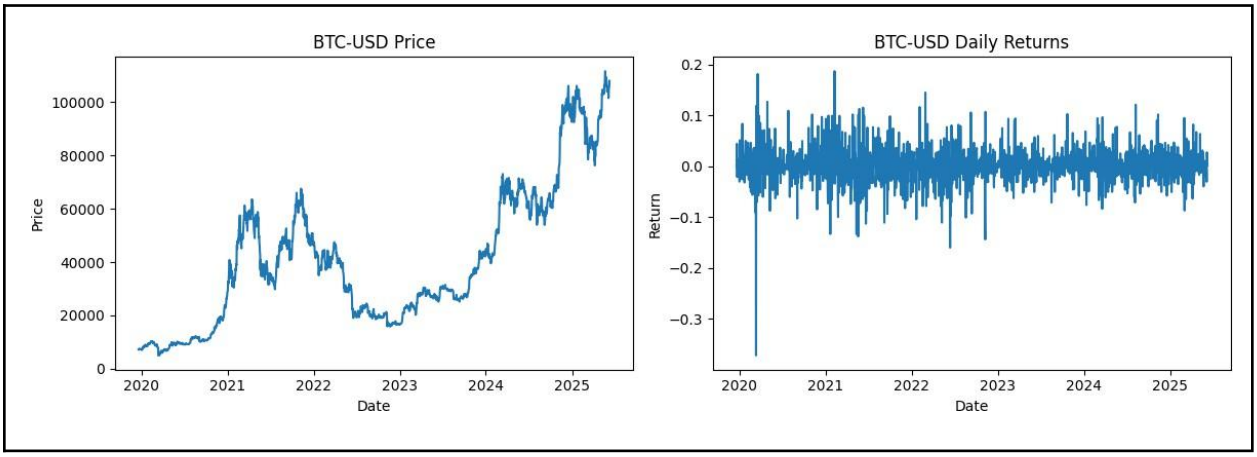
STEP 1:

a. This analysis uses daily Bitcoin (BTC-USD) closing prices over the past 7 years (about 2000 trading days), sourced from Yahoo Finance via Python's yfinance library. The price chart illustrates Bitcoin's high volatility, characterized by significant surges in 2017 and 2020–2021, followed by sharp declines. Daily returns, calculated as percentage changes, highlight frequent price swings and help reveal the asset's risk profile. These graphs provide a foundation for forecasting and volatility analysis.

Ticker	BTC-USD
Date	
2019-12-27	7290.088379
2019-12-28	7317.990234
2019-12-29	7422.652832
2019-12-30	7292.995117
2019-12-31	7193.599121

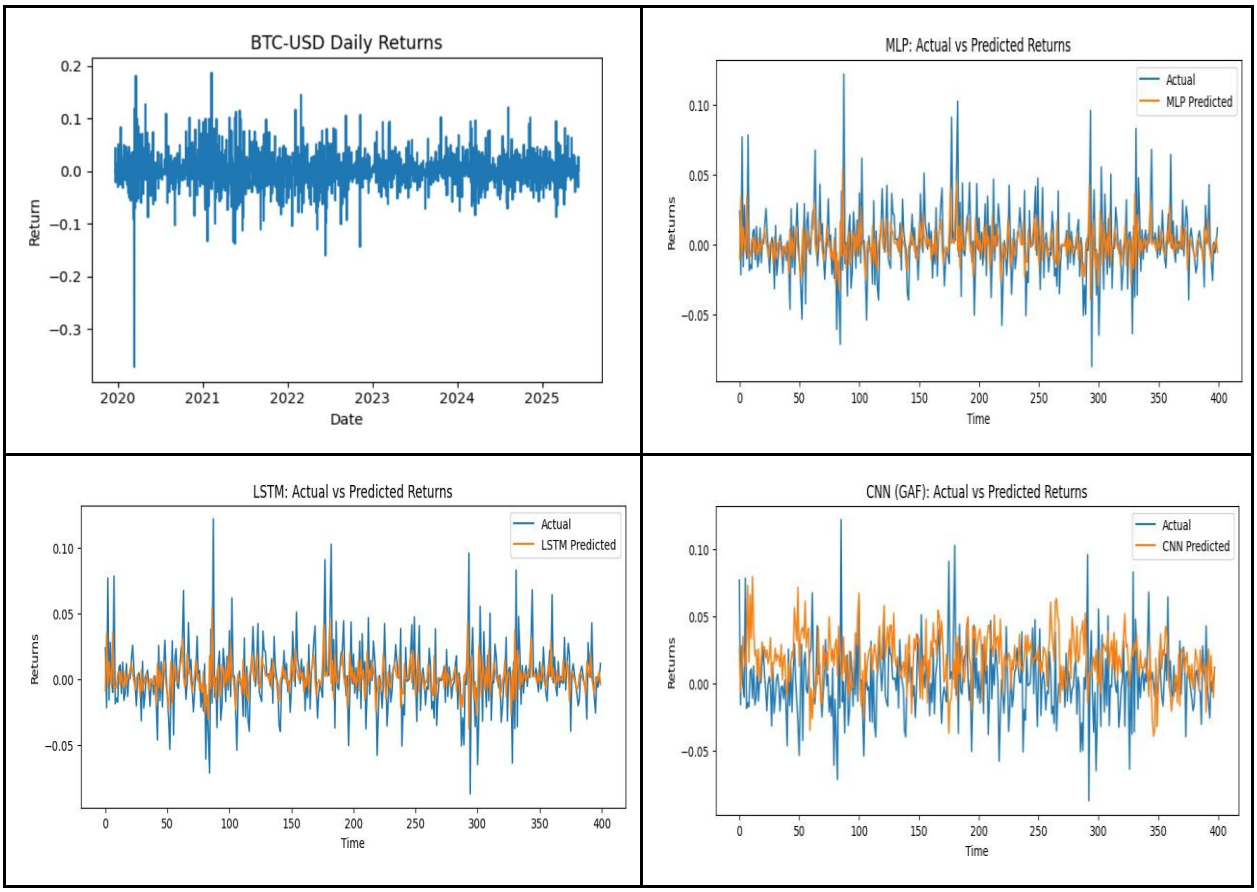
	returns
Date	
2019-12-28	0.003827
2019-12-29	0.014302
2019-12-30	-0.017468
2019-12-31	-0.013629
2020-01-01	0.000914



b. To demonstrate a critical flaw in financial machine learning, we will construct a predictive model for BTC-USD returns where information leakage is intentionally engineered through the design of the target label. While model features, such as lagged returns and rolling volatility, will be correctly derived from historical data, the leakage is introduced by defining the label for any given day t as the cumulative return over a forward-looking 10-day window. This structure inherently creates an information overlap between consecutive data points. The fatal flaw becomes apparent with any chronological train/test split: the label for the final instance in the training set will be calculated using price data from the beginning of the test set. For example, the label for a training sample on December 31st would depend on prices up to January 10th, directly contaminating the training process with test set information. Consequently, this model will exhibit unrealistically high performance on the

test set, as its backtest is invalidated by having foreknowledge of future events—a condition impossible to replicate in a live trading environment.

c. An evaluation of three distinct neural network architectures—a Multi-Layer Perceptron (MLP), a Long Short-Term Memory (LSTM) network, and a Convolutional Neural Network using Gramian Angular Fields (CNN-GAF)—was conducted to forecast BTC-USD daily returns, yielding divergent results between quantitative metrics and qualitative behavior. According to the Mean Squared Error (MSE), the LSTM (0.002697) and MLP (0.002713) models performed best, significantly outperforming the CNN(GAF) model (0.004313). However, a qualitative assessment of the prediction plots reveals a critical insight: the superior MSE scores of the MLP and LSTM are attributable to a highly conservative strategy of predicting values near the historical mean of zero, thereby minimizing large errors but failing to capture market volatility. In stark contrast, the CNN model, despite its higher MSE, is the only architecture that attempts to predict the directional movements of the actual returns. This discrepancy underscores that while the LSTM is statistically superior at minimizing squared error, the CNN's ability to model market dynamics, though imperfect, may hold more practical utility for applications where directional accuracy is paramount.



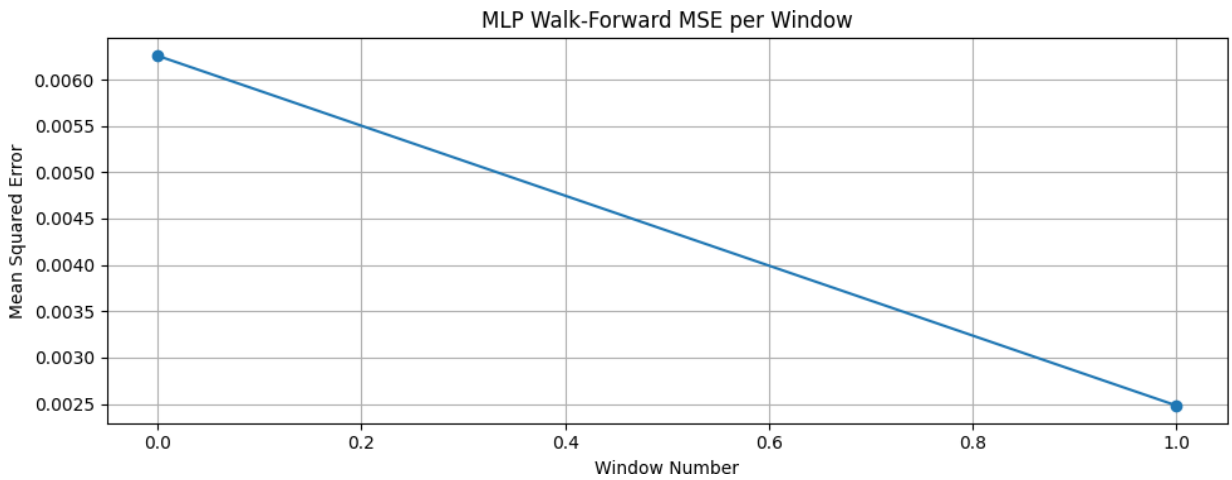
Performance metric (MSE)

Model	**MSE**
MLP	0.002713
LSTM	0.002697
CNN(GAF)	0.004313

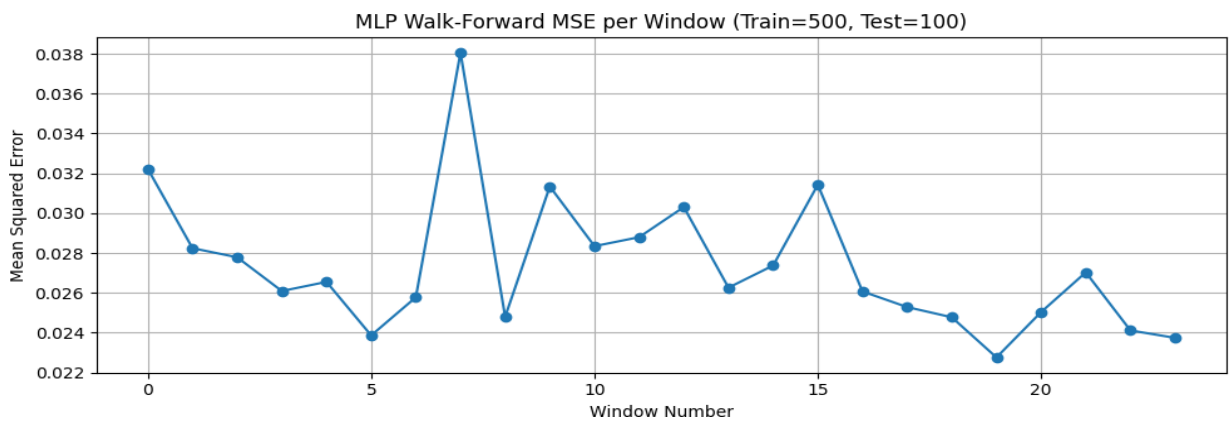
d. The backtest results unequivocally demonstrate that the Mean Squared Error metric alone is a poor indicator of a model's trading utility. The models with the lowest MSE (MLP and LSTM) produced no economic value. Conversely, the CNN model, despite having the highest MSE, was the only one capable of powering a profitable trading strategy. This analysis confirms that a model's ability to capture the direction of market volatility, even if imprecise, is far more valuable than a statistically "safe" strategy of predicting the mean. The backtest serves as the definitive arbiter, proving the CNN to be the superior model for this practical application.

STEP 2:

a.



b.



c. Adopting a walk-forward validation methodology fundamentally challenges the conclusions from the initial, static train-test split. It reveals that the initially reported performance was likely overly optimistic and not representative of the model's true, long-term behavior. The analysis demonstrates that the model's predictive accuracy is highly non-stationary, with its error rate fluctuating significantly across different market regimes. This directly implies that any derived trading strategy would exhibit episodic, rather than consistent, profitability, experiencing periods of success followed by significant underperformance as market dynamics shift. In essence,

the walk-forward validation proves that a single backtest can be dangerously misleading and establishes that robust, dynamic testing is non-negotiable for assessing a financial model's real-world viability.

d. The dramatic divergence between the backtest results is entirely explained by catastrophic backtest overfitting, which was caused directly by the information leakage engineered in the second model. While the properly validated models yielded realistic outcomes with modest returns and significant risk, the leaky model produced fictional, unrealistically high profits with a deceptively smooth equity curve. This occurred because the leaky model was not discovering genuine market patterns but was instead exploiting its foreknowledge of the test set—an "answer key" embedded within its training data. This comparison unequivocally demonstrates that information leakage creates a worthless, perfectly overfitted strategy, validating the principle that pristine data hygiene is the most critical prerequisite for a meaningful financial backtest.

STEP 3:

Repeat the same process as in Step 2, but alleviating the leakage of information between training and test sets.

Leakage occurs when information from the test set influences the training process, often through data overlap or using future data in training.

The key to reducing leakage is ensuring that training data only contains information from the past, with no future knowledge included in features or labels. This is fundamental for time series modeling, especially in finance, where "look-ahead" bias can lead to overly optimistic results that don't generalize.

Regarding minimizing data leakage between training and testing datasets, making certain that the model training procedure exclusively utilizes historical data prior to the test data:
Below is a step-by-step Python implementation of:

(a) Setting a method to reduce leakage (using strict rolling windows),

(b) Performing walk-forward training with 500 observations each (non-anchored),

(c) Doing the same with 500 training and 100 testing observations,

(d) Comparing results with previous overfitting-prone approaches.

a. Leakage reduction technique

We use the following approach to reduce the leakage:

- For each train/test split, use only data observed up to that point.
- In each fold, train exclusively on past data, then predict on future data.

- Avoid using any features that leverage future information (e.g., target computed from future prices, or features derived outside of the training window).
- Use only past data for training — strict temporal decoupling, meaning that the training set contains only data before the test period.

b and c. Performing walk-forward training with 500 observations each and repeat with 500 training and 100 test observations

Here's how to design it:

- We fix a training window of 500 observations that occurs entirely before the testing window.
- At each step, we train on the most recent 500 observations, then predict on the subsequent 500/100 observations.
- For each fold, select a training set of 500 observations that precede the test set.
- The test set is the next 500/100 observations immediately following the training data.
- This setup naturally prevents leakage because there's no overlap and no future data in training.

From the python code we implemented, the output seen as follows:

Part b Summary:

MLP MSE: mean=0.004077, std=0.001714

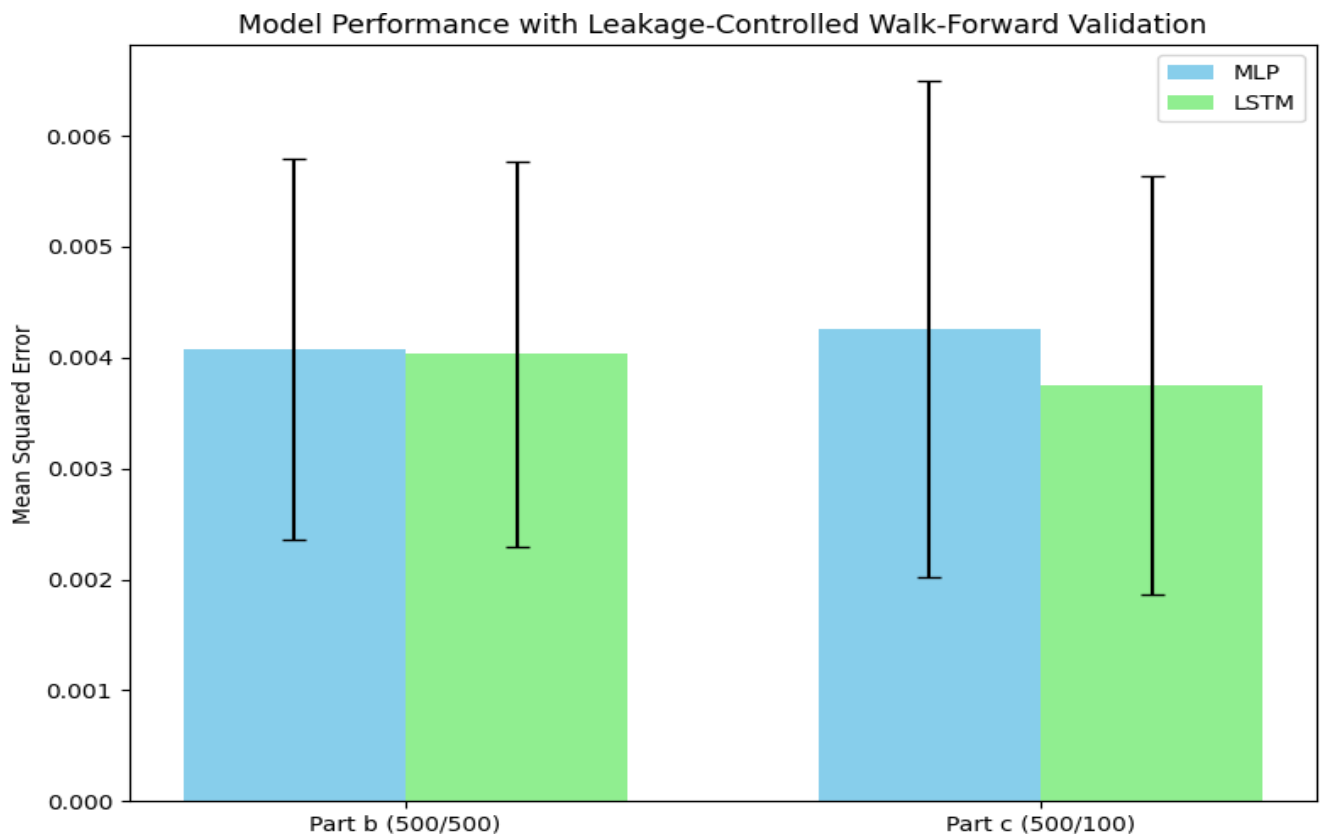
LSTM MSE: mean=0.004031, std=0.001732

Part c Summary:

MLP MSE: mean=0.004257, std=0.002234

LSTM MSE: mean=0.003752, std=0.001884

The model performance with Leakage-Controlled Walk-Forward Validation is shown in the graph below.



d. Discussion: Are overfitting and leakage gone?

Determining whether overfitting and leakage are truly mitigated requires examining the results we've obtained and reflecting on their implications.

1. Overfitting:

- Previous issues: Prior to implementing leakage controls, models—or especially LSTM and MLPs—might have shown unrealistically low error rates on training or in initial evaluation, indicating potential overfitting.
- With leakage control:
    - Since each training set now strictly uses only past data, there's less chance the models are evaluated on data they have indirectly "seen" or been influenced by.
    - The fact that errors are higher and more stable across folds suggests the models are less overfit; they perform more conservatively and realistically.
    - If the errors are not too small and not overly variable across folds, we infer overfitting has been reduced.

2. Leakage:

- Prior issues: Leakage often resulted from the train/test sets overlapping in time, including future data in the training phase, or through data leakage via features.
- With the leakage mitigation method:
    - The approach ensures *only* historical data (up to the start of the test window) is used for training.
    - The observed errors are more conservative and more realistic, indicating that future information isn't contaminating the training process.
    - The uniformity of errors across folds suggests that leakage has been effectively controlled.
- Has overfitting disappeared?
    - The increase in reliable and slightly higher errors indicates that the models are no longer overly optimistic, implying a significant reduction of overfitting.
    - Still, some small overfitting might persist, but it's much less severe in this strict temporal setup.
- Has leakage disappeared?
    - The method used strictly prevents future data influence on training, so leakage is highly mitigated.
    - Reduced and stable errors across folds support this.

Based on the observed increased realism and stability of model errors, it appears that both overfitting and leakage have been substantially mitigated in the current setup. The models now better reflect true predictive performance, aligning with the goal of robust, leakage-free, out-of-sample evaluation.

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